SIMERJEET SINGH

New Delhi, India +91-7840844961 simerjeetsingh20232027@gmail.com

Portfolio LinkedIn GitHub LeetCode

PROFESSIONAL SUMMARY

Motivated Computer Science undergraduate at MSIT, New Delhi, with strong full-stack development foundations in Next.js, React, and Node.js, and hands-on AI/LLM integration experience using Groq, Gemini Embeddings, and RAG pipelines. Passionate about building scalable production-ready systems. Actively seeking opportunities to apply software engineering principles and emerging AI capabilities to solve real-world problems.

SKILLS

- **Programming** — TypeScript, JavaScript, Python
- **Full-Stack Development** — Next.js, React, Node.js, Tailwind CSS, Prisma ORM, REST API design; system design fundamentals
- **Databases & Storage** — PostgreSQL (pgvector), MongoDB, Redis; vector similarity search, session state management, schema design
- **AI / LLM Integration** — Groq API (Llama 3.3), Gemini Embeddings, RAG pipelines, text chunking, semantic search
- **Cloud & DevOps** — Docker, Vercel, Render; CI/CD basics, cloud deployment workflows, environment configuration
- **Testing & Tools** — Git/GitHub, Clerk Auth, Socket.io; unit testing fundamentals, debugging, performance profiling

PROJECTS

➤ **AudienceAI** (GitHub | Live Demo) – AI-powered live presentation Q&A platform enabling real-time, anonymous audience engagement at scale.

- Developed a production-ready platform using Next.js, TypeScript, and Socket.io, enabling instant, anonymous audience engagement via QR code — no account creation required.
- Integrated Groq's Llama 3.3 model to automatically cluster semantically similar questions and generate context-aware answers for presenters in real time, reducing repetitive Q&A overhead.
- Architected Redis-backed session state management to sustain 1000+ concurrent users with low-latency Socket.io delivery, demonstrating scalable system design principles.
- Implemented Clerk Auth for secure, role-based presenter dashboards with full moderation control; deployed on Render.

➤ **DocuMind** (GitHub | Live Demo) – AI document intelligence platform allowing users to upload PDFs and query content through a full RAG pipeline with cited, LLM-generated responses.

- Engineered an end-to-end RAG pipeline covering PDF ingestion, intelligent text chunking, Gemini embedding generation, and pgvector-powered semantic similarity search on PostgreSQL.
- Delivered context-aware document Q&A using Groq LLM with cited source passages, enabling answer traceability and improving user trust in AI-generated responses.
- Designed multi-tenant architecture with Prisma ORM and Clerk Auth to enforce strict user-specific document isolation, applying software engineering best practices for data integrity.
- Engineered scalable multi-tenant architecture with robust user-specific document isolation, leveraging pgvector for efficient semantic search and embedding workflows across large document sets.

➤ **AI DSA Visualiser** (GitHub | Live Demo) – Interactive educational platform visualizing Data Structures & Algorithms with AI-assisted difficulty analysis.

- Developed an interactive educational platform integrating complex algorithmic logic with modern frontend frameworks and AI-driven features to enhance learning outcomes for DSA learners.
- Built step-by-step visual simulations for sorting, searching, graph traversal, and dynamic programming in React, with real-time highlighting of active states and data transformations.
- Implemented an AI layer assigning difficulty ratings and estimated solve times to 50+ problems, enabling users to build structured, progressive practice roadmaps.
- Deployed on Vercel with optimized build pipeline; applied component-based system design to ensure modularity and easy addition of new algorithm modules.

CODING PLATFORMS

- Solved 400+ problems across platforms including LeetCode and GeeksforGeeks, covering data structures algorithms.

EDUCATION

Maharaja Surajmal Institute of Technology — Aug 2023 – Jun 2027
B.Tech in Computer Science and Engineering — CGPA: 8.47 (up to 5th Sem)